

TSIA2 Math Practice Test 11

PART A — QUESTIONS

TSIA2-M11-001

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A student drives 36 miles round trip to campus each day for 5 days. The car averages 30 miles per gallon, and gasoline costs \$3.20 per gallon. What is the student's gasoline cost for the 5 days?

A) \$17.28

B) \$19.20

C) \$24.00

D) \$30.00

TSIA2-M11-002

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve for x: $3(2x - 1) + 4 = 5x + 12$

A) 9

B) 10

C) 11

D) 13

TSIA2-M11-003

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular piece of carpet is 12 feet long and 9 feet wide. A 3-foot by 4-foot rectangular section is cut out of one corner. How many square feet of carpet remain?

A) 84 square feet

B) 96 square feet

C) 105 square feet

D) 108 square feet

TSIA2-M11-004

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A survey asked 60 students how they usually travel to campus.

Bus: 18 students

Bike: 12 students

Walk: 10 students

Car: 20 students

If one surveyed student is chosen at random, what is the probability that the student usually bikes or walks to campus?

- A) $\frac{1}{3}$
- B) $\frac{11}{30}$
- C) $\frac{2}{5}$
- D) $\frac{11}{20}$

TSIA2-M11-005

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

At an orientation event, the recommended ratio is 5 mentors for every 18 students. If 126 students attend, how many mentors are recommended?

- A) 30
- B) 35
- C) 40
- D) 45

TSIA2-M11-006

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $(2a - 5) - 3(a + 4) + 7a$?

- A) $6a - 17$
- B) $6a + 7$
- C) $12a - 17$
- D) $12a + 7$

TSIA2-M11-007

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

An aquarium is shaped like a rectangular prism. It is 24 inches long, 10 inches wide, and 12 inches high. If it is filled to $\frac{3}{4}$ of its full volume, how many cubic inches of water does it contain?

- A) 1,440 cubic inches
- B) 1,800 cubic inches
- C) 2,160 cubic inches
- D) 2,880 cubic inches

TSIA2-M11-008

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

Two groups took the same quiz.

Group A scores: 70, 75, 80, 85, 90

Group B scores: 60, 75, 80, 85, 100

How much greater is the range of Group B's scores than the range of Group A's scores?

- A) 10 points
- B) 15 points
- C) 20 points
- D) 30 points

TSIA2-M11-009

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A lab procedure uses 0.75 teaspoon of powder for every 8 ounces of water. How many teaspoons of powder are needed for 40 ounces of water?

- A) 3 teaspoons
- B) $3\frac{1}{2}$ teaspoons
- C) $3\frac{3}{4}$ teaspoons
- D) $4\frac{1}{4}$ teaspoons

TSIA2-M11-010

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve the system of equations:

$$4x + y = 23$$

$$x + y = 8$$

What is the value of y ?

- A) 2
- B) 3
- C) 5
- D) 8

TSIA2-M11-011

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A circular track has a circumference of 40π meters. What is the area enclosed by the track?

- A) 100π square meters
- B) 200π square meters
- C) 400π square meters
- D) $1,600\pi$ square meters

TSIA2-M11-012

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A spinner has 6 equal sections numbered 1 through 6. A coin is flipped once. What is the

probability that the spinner lands on an even number and the coin lands heads?

- A) $\frac{1}{6}$
- B) $\frac{1}{4}$
- C) $\frac{1}{3}$
- D) $\frac{1}{2}$

TSIA2-M11-013

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A warehouse inventory decreased from 1,050 items to 870 items over 6 weeks. What was the average decrease in inventory per week?

- A) 25 items per week
- B) 30 items per week
- C) 35 items per week
- D) 45 items per week

TSIA2-M11-014

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A student has 35 minutes available to complete online practice modules. Each module takes 18 minutes. What is the greatest number of complete modules the student can finish in at most 120 total minutes of study time?

- A) 3
- B) 4
- C) 5
- D) 6

TSIA2-M11-015

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A 6-foot-tall person casts a 4.5-foot shadow at the same time a flagpole casts an 18-foot shadow. Assuming the triangles formed are similar, how tall is the flagpole?

- A) 13.5 feet
- B) 21 feet
- C) 24 feet
- D) 27 feet

TSIA2-M11-016

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

In a sample of 50 students, 32 reported having a part-time job. A counselor claims that more than 60% of the sampled students have a part-time job. Which statement is supported by the data?

- A) The claim is supported because $\frac{32}{50} = 64\%$, which is greater than 60%.

- B) The claim is supported because $32/50 = 60\%$, which is equal to 60%.
- C) The claim is not supported because $32/50 = 58\%$, which is less than 60%.
- D) The claim is not supported because fewer than 32 students would be needed.

TSIA2-M11-017

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

An annual online subscription normally costs \$96. A new customer pays $3/4$ of the regular annual price. What is the equivalent monthly cost for the new customer?

- A) \$6
- B) \$8
- C) \$12
- D) \$72

TSIA2-M11-018

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A bacteria culture starts with 150 bacteria and doubles every hour. Which expression represents the number of bacteria after h hours?

- A) $150 + 2h$
- B) $150h + 2$
- C) $150 \cdot 2^h$
- D) $2 \cdot 150^h$

TSIA2-M11-019

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

Two angles are complementary. One angle measures 28° . What is the measure of the other angle?

- A) 52°
- B) 62°
- C) 118°
- D) 152°

TSIA2-M11-020

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A course grade is based 70% on exams and 30% on assignments. A student's exam average is 84, and the assignment average is 92. What is the student's overall course average?

- A) 84.8
- B) 86.4
- C) 88.0
- D) 89.6

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M11-001 — Correct: B — Correct Answer: \$19.20 — Explanation: The student drives $36 \times 5 = 180$ miles. At 30 miles per gallon, the car uses $180 \div 30 = 6$ gallons. The cost is $6 \times \$3.20 = \19.20 .

TSIA2-M11-002 — Correct: C — Correct Answer: 11 — Explanation: Distribute and simplify: $6x - 3 + 4 = 5x + 12$, so $6x + 1 = 5x + 12$. Subtract $5x$ and 1 to get $x = 11$.

TSIA2-M11-003 — Correct: B — Correct Answer: 96 square feet — Explanation: The full carpet area is $12 \times 9 = 108$ square feet. The cut-out section has area $3 \times 4 = 12$ square feet, so $108 - 12 = 96$ square feet remain.

TSIA2-M11-004 — Correct: B — Correct Answer: 11/30 — Explanation: A total of $12 + 10 = 22$ students bike or walk. Out of 60 students, the probability is $22/60 = 11/30$.

TSIA2-M11-005 — Correct: B — Correct Answer: 35 — Explanation: Since $126 \div 18 = 7$, the event has 7 groups of 18 students. At 5 mentors per group, $7 \times 5 = 35$ mentors are recommended.

TSIA2-M11-006 — Correct: A — Correct Answer: $6a - 17$ — Explanation: Distribute and combine like terms: $2a - 5 - 3a - 12 + 7a = 6a - 17$.

TSIA2-M11-007 — Correct: C — Correct Answer: 2,160 cubic inches — Explanation: The full volume is $24 \times 10 \times 12 = 2,880$ cubic inches. Three-fourths of 2,880 is 2,160 cubic inches.

TSIA2-M11-008 — Correct: C — Correct Answer: 20 points — Explanation: Group A's range is $90 - 70 = 20$. Group B's range is $100 - 60 = 40$. The difference is $40 - 20 = 20$ points.

TSIA2-M11-009 — Correct: C — Correct Answer: $3 \frac{3}{4}$ teaspoons — Explanation: Since 40 ounces is 5 times 8 ounces, multiply 0.75 by 5: $0.75 \times 5 = 3.75$, or $3 \frac{3}{4}$ teaspoons.

TSIA2-M11-010 — Correct: B — Correct Answer: 3 — Explanation: Subtract $x + y = 8$ from $4x + y = 23$ to get $3x = 15$, so $x = 5$. Then $5 + y = 8$, so $y = 3$.

TSIA2-M11-011 — Correct: C — Correct Answer: 400π square meters — Explanation: Since $C = 2\pi r$ and $40\pi = 2\pi r$, the radius is 20 meters. The area is $\pi r^2 = \pi(20^2) = 400\pi$ square meters.

TSIA2-M11-012 — Correct: B — Correct Answer: $\frac{1}{4}$ — Explanation: The probability of spinning an even number is $\frac{3}{6} = \frac{1}{2}$, and the probability of heads is $\frac{1}{2}$. The combined probability is $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

TSIA2-M11-013 — Correct: B — Correct Answer: 30 items per week — Explanation: The inventory decreased by $1,050 - 870 = 180$ items. Over 6 weeks, the average decrease was $180 \div 6 = 30$ items per week.

TSIA2-M11-014 — Correct: B — Correct Answer: 4 — Explanation: The available remaining study time is $120 - 35 = 85$ minutes. Since $85 \div 18$ is about 4.72, the student can complete at most 4 full modules.

TSIA2-M11-015 — Correct: C — Correct Answer: 24 feet — Explanation: Set up the proportion $x/18 = 6/4.5$. Then $x = 18 \times 6 \div 4.5 = 24$ feet.

TSIA2-M11-016 — Correct: A — Correct Answer: The claim is supported because $32/50 = 64\%$, which is greater than 60%. — Explanation: The sample proportion is $32 \div 50 = 0.64$, or 64%, so the counselor's claim is supported.

TSIA2-M11-017 — Correct: A — Correct Answer: \$6 — Explanation: The new customer pays $\frac{3}{4}$ of \$96, which is \$72. Dividing \$72 by 12 months gives \$6 per month.

TSIA2-M11-018 — Correct: C — Correct Answer: $150 \cdot 2^h$ — Explanation: Doubling each hour means multiplying by 2 once for each hour. Starting with 150 bacteria, the expression after h hours is $150 \cdot 2^h$.

TSIA2-M11-019 — Correct: B — Correct Answer: 62° — Explanation: Complementary angles have a sum of 90° . The other angle is $90^\circ - 28^\circ = 62^\circ$.

TSIA2-M11-020 — Correct: B — Correct Answer: 86.4 — Explanation: Compute the weighted average: $0.70(84) + 0.30(92) = 58.8 + 27.6 = 86.4$.